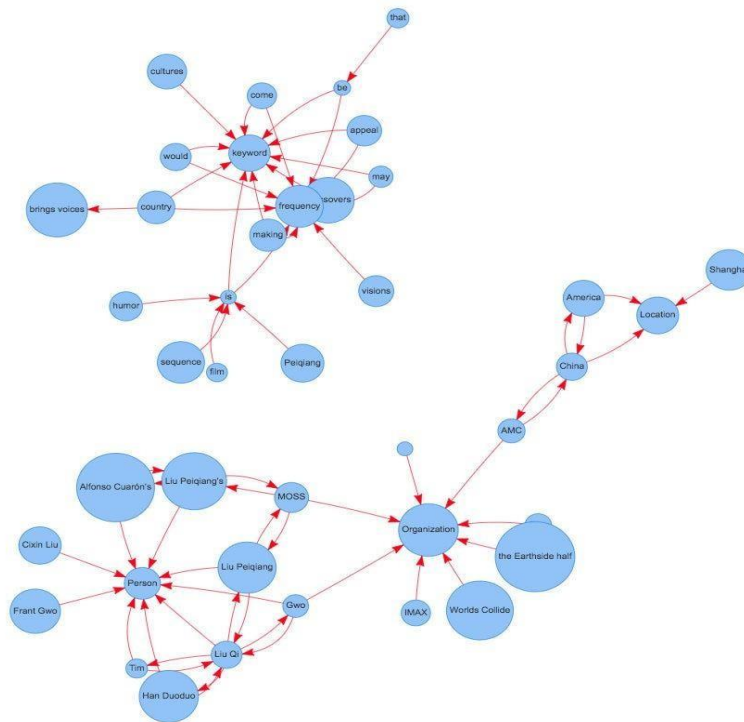


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Aim: TextRank for Document Summarization

IDE: Google Colab **Theory:**

TextRank is an algorithm based on PageRank, which often used in keyword extraction and text summarization. In this article, I will help you understand how TextRank works with a keyword extraction example and show the implementation by Python.



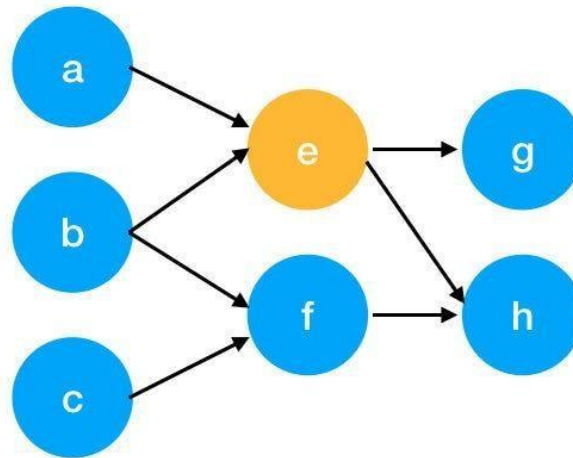
Understand PageRank

There are tons of articles talking about PageRank, so I just give a brief introduction to PageRank. This will help us understand TextRank later because it is based on PageRank. PageRank (PR) is an algorithm used to calculate the weight for web pages. We can take all web pages as a big directed graph. In this graph, a node is a webpage. If webpage A has the link to web page B, it can be represented as a directed edge from A to B. After we construct the whole graph, we can assign weights for web pages by the following formula.

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$$S(V_i) = (1 - d) + d * \sum_{j \in In(v_i)} \frac{1}{|Out(V_j)|} S(V_j)$$

- $S(V_i)$ - the weight of webpage i
- d - damping factor, in case of no outgoing links
- $In(V_i)$ - inbound links of i, which is a set
- $Out(V_j)$ - outgoing links of j, which is a set
- $|Out(V_j)|$ - the number of outbound links



Here is an example to better understand the notation above. We have a graph to represent how web pages link to each other. Each node represents a webpage, and the arrows represent edges. We want to get the weight of webpage e. We can rewrite the summation part in the above function to a simpler version.

$$\begin{aligned}
 In(v_e) &= \{a, b\}, j \in \{a, b\} \\
 \sum_{j \in \{a, b\}} \frac{1}{|Out(V_j)|} S(V_j) &= \frac{1}{|Out(V_a)|} S(V_a) + \frac{1}{|Out(V_b)|} S(V_b) \\
 &= \frac{1}{|\{e\}|} S(V_a) + \frac{1}{|\{e, f\}|} S(V_b) \\
 &= S(V_a) + \frac{1}{2} S(V_b)
 \end{aligned}$$

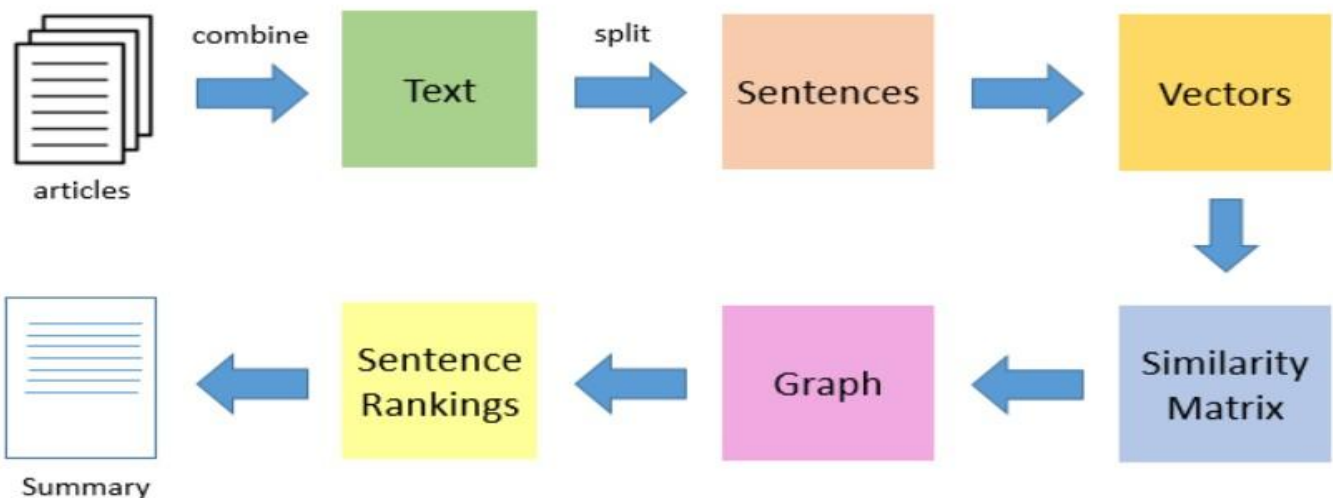
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We can get the weight of webpage e by the following function.

$$S(V_e) = (1 - d) + d * \left(S(V_a) + \frac{1}{2} S(V_b) \right)$$

We can see the weight of the webpage e is dependent on the weights of its inbound pages. We need to run this iteration much time to get the final weight. In the initialization, the importance of each webpage is 1.

TextRank for document Summarization



TextRank works in the following steps:

1. Tokenize documents into sentences.
2. Preprocess each sentence in the document.
3. Count key phrases and normalize them or produce TFIDF Matrix, you can also use any kind of vectorization such as spacy vectors.
4. Calculate the Jaccard Similarity between sentences and key phrases.
5. Rank the sentences with higher significance.

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Pre Lab Exercise:

1. How keyword extraction is similar to document summarization process?

Keyword extraction and document summarization are similar because both aim to reduce large text into a shorter and more meaningful form. In keyword extraction, we find important words, while in summarization, we select important sentences. Both methods focus on identifying the most relevant information from the text using techniques like ranking and scoring (such as TextRank based on PageRank).

2. How keyword extraction is different to document summarization process?

The main difference is in the output. Keyword extraction gives only important words or phrases, whereas document summarization provides meaningful sentences or a paragraph. Keywords do not explain the full idea, but summaries give a clear understanding of the text. Also, summarization considers sentence relationships, while keyword extraction focuses on word importance.

3. Limitation of TextRank as a summarizer

TextRank has some limitations:

- It does not understand the actual meaning of text (no deep semantic understanding).
- It may ignore important sentences if they are not strongly connected in the graph.
- It depends heavily on word/sentence frequency and structure.
- Sometimes the generated summary may not be fully coherent or smooth.

Program (Code):

```
import nltk
import networkx as nx
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
import string

nltk.download('punkt')
nltk.download('stopwords')
nltk.download('punkt_tab') # Added to resolve LookupError
```

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```
def textrank_keywords(text, window_size=4, top_n=5):
```

```
    words = word_tokenize(text.lower())
```

```
    stop_words = set(stopwords.words('english'))
```

```
    words = [w for w in words if w not in stop_words and w not in string.punctuation]
```

```
    graph = nx.Graph()
```

```
    for i, word in enumerate(words):
```

```
        for j in range(i+1, i+window_size):
```

```
            if j < len(words):
```

```
                graph.add_edge(word, words[j])
```

```
    scores = nx.pagerank(graph)
```

```
    ranked_words = sorted(scores.items(), key=lambda x: x[1], reverse=True)
```

```
    keywords = [word for word, score in ranked_words[:top_n]]
```

```
    return keywords
```

```
text = """
```

I am a student of Information and Communication Technology with a strong interest in programming and software development. I enjoy working on practical projects like Flutter apps, SQL databases, and microcontroller-based systems. I have experience in languages like C, Python, and Dart, and I like building applications such as Todo apps and expense trackers.

I am also interested in Artificial Intelligence and Machine Learning, where I explore concepts like TextRank, data visualization, and probability. I prefer learning by doing practical experiments and coding implementations. My goal is to improve my technical skills and build useful real-world applications.

```
"""
```

```
keywords = textrank_keywords(text)
```

```
print("Extracted Keywords:", keywords)
```

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Results:

• Extracted Keywords: ['like', 'practical', 'apps', 'learning', 'applications']

Observation:

The TextRank algorithm works effectively for extracting important keywords without supervision. It uses graph-based ranking to identify important words. However, it does not understand context deeply, so sometimes results may miss meaningful information. Overall, it is simple, fast, and useful for basic NLP tasks.

Post Lab Exercise:

1. Write about “your ownself” (in not more than 500 words ⑦ You know better about you, rather than ChatGPT!!). Generate the summary of your portfolio in
 - a. 50% of the size of your portfolio
 - b. 25% of the size of your portfolio

Comment over the summary obtained. Which sentence you think should be there in your summary, but was not spotted by textrank? Rate the overall summary in the scale of 1-5 (with 1 as least and 5 as highest rating). Paste the code, your portfolio, output and your analysis.

I am a student of Information and Communication Technology with a strong interest in programming and software development. I enjoy working on practical projects like Flutter apps, SQL databases, and microcontroller-based systems. I have experience in languages like C, Python, and Dart, and I like building applications such as Todo apps and expense trackers. I am also interested in Artificial Intelligence and Machine Learning, where I explore concepts like TextRank, data visualization, and probability. I prefer learning by doing practical experiments and coding implementations. My goal is to improve my technical skills and build useful real-world applications.